

04 — Low-Level Design (Stage 4): index

Version	0.1
Date	2026-09-24
Based on	BRD v0.6.1, PRD v0.6, HLD v0.2 (all approved)
Status	AWAITING APPROVAL

Contents

Deliverable	File
Shared conventions and reconciliation decisions R-1...R-22	conventions.md
Event contract: 68 events (AsyncAPI 3.0, generated)	../contracts/asyncapi/events.yaml · table: events-table.md · generator: <code>tools/gen_events.py</code>
API contracts: 240 operations (OpenAPI 3.1)	<code>contracts/openapi/{web,intake,records,journeys,crm-engine,listings,insight}.yaml</code> + <code>_common.yaml</code> · table: endpoints-table.md · generator: <code>tools/gen_gate_tables.py</code>
Per-service LLDs (schema, indexes with their queries, rules, events, errors, PII, retention, module layout)	web · intake · records · journeys · crm-engine · listings · insight
Data hosting	data-hosting.md
Capacity plan and load test	capacity-plan.md

Verification done

- All 8 OpenAPI files and the AsyncAPI file parse. The OpenAPI files pass OpenAPI 3.1 validation (checked by the drafting agents).
- All **240 operations** declare security, roles, rate limit, timeout, idempotency, emitted events and RFC 7807 errors. Lists use cursor pagination (max 100; 50 on the public Listings API).
- Every emitted or consumed event exists in the catalogue. Every event appears in its producer's and each consumer's LLD.
- Every table index is listed with the query it serves (LLD §3 of each service).

Decisions taken during Stage 4 (for your awareness)

The conventions (§10) record R-1...R-22. The ones that touch product behaviour:

- **R-8:** a photo with text on it is a warning, not a block (as the PRD says).
- **R-9:** records from an unknown city count as Mumbai when the ad came from a Mumbai edition. Otherwise they go to review.
- **R-10:** extractor rows start at "Enriched". Quick add starts at "Captured", or "Contacted" if entered during a call.
- **R-11:** controlled values are matched ignoring case and spaces. Inverted ranges (min > max) are rejected.
- **R-12:** a demand open to several deal types ages on the shortest life curve. A retired offer can be revived by a confirmed call.
- **R-14:** an unknown area basis gets $\pm 25\%$ tolerance and a flag, with no conversion between carpet and built-up.

- **R-16:** exports are capped at 20,000 rows in the pilot and 100,000 in production.
- **R-17:** chat conversations are private to their author. Actions taken through chat are audit-logged.
- Contact details are **masked by default** in the app. Showing a full phone number or unit is an explicit, audit-logged "reveal" (records `POST /v1/reveals`, 60 per hour per user).
- **Staff edits win:** a later extractor upload never overwrites a field someone changed by hand.
- A split record keeps its CRM work on the first child when the extractor re-splits an ad.

Questions for the product owner

#	Question	Default if you don't mind
Q4-1	2-step sign-in: should staff use Google Workspace accounts (so 2-step verification can be enforced), or personal Gmail with 2-step as a policy only?	Google Workspace for 11 Estates staff
Q4-2	May Data operators export contact columns?	No (Admins, Managers and agents only)
Q4-3	Proposal feedback options: liked / rejected / visit requested. Add "maybe"?	Add "maybe"
Q4-4	Fake phone format for the pilot anonymiser: <code>+91 00000 xxxxx</code> (never a real number)	As proposed

Items to verify at provisioning (Stage 5 foundation tasks, from data-hosting §7)

- Supabase standby/failover versus the 99.9% target (may need a CR).
- The connection limit per compute size.
- `pgmq`, `pg_cron` and `pg_net` on the Free plan in Mumbai.
- The Hugging Face endpoint region.

Minor open items (don't block approval; resolved during implementation)

- A person-purge event for retention (`person.purged.v1`, journeys G-6).
- The developer name on public projects is now in `project.*` events.
- Digi (lead-form) exports in mapping mode need extra optional fields (campaign/listing refs, photo URLs): intake Q-I3.
- The keys for demand must-haves (`parking`, `amenity:<name>`): crm-engine C-10.
- `publicDescriptionSource` is not used in Phase 1. Public descriptions are generated from fields (listings OQ-L4).

Event catalogue (generated)

Delivery: at least once; ordering per aggregate; dedupe on eventId. Source: `contracts/asynccapi/events.yaml`.

Event	Producer	Consumers	Meaning
upload.started.v1	intake	insight	An upload began processing.
rows.classified.v1	intake	records, insight	A batch (≤ 500) of classified rows is ready. Carries no PII: records pulls contact fields via intake GET /internal/v1/uploads/{uploadId}/rows?batch=.
upload.completed.v1	intake	web, insight	All batches of an upload were emitted.
upload.failed.v1	intake	web, insight	An upload could not be processed.
review_item.created.v1	intake	insight	A row needs classification review.
review_item.resolved.v1	intake	records, insight	A classification review item was resolved by a person.
offer.created.v1	records	journeys, crm-engine, listings, insight	A new Offer exists (from intake, quick add, Add supply, price sheet or renewal).
offer.updated.v1	records	journeys, crm-engine, listings, insight	Offer or its Property facts changed (full current facts).
offer.price_changed.v1	records	journeys, crm-engine, listings, insight	Price fields changed.
offer.record_stage_changed.v1	records	journeys, listings, insight	Record axis moved (e.g. Contacted → Verified).
demand.created.v1	records	journeys, crm-engine, listings, insight	A new Demand exists.
demand.updated.v1	records	journeys, crm-engine, listings, insight	Demand facts changed (full current facts).
demand.touch_added.v1	records	journeys, insight	Another arrival of the same demand (first touch unchanged).
enquiry.received.v1	records	journeys, insight	A market enquiry linked to an offer/project/campaign.
records.merged.v1	records	journeys, crm-engine, listings, insight	Two records were merged (reversible).
records.merge_undone.v1	records	journeys, crm-engine, listings, insight	A merge was undone.
person.flagged.v1	records	journeys, insight	A person got a flag (invalid, broker-posing, unwilling, anonymous-shares-only).
watchlist_item.created.v1	records	journeys, insight	A Market Signal was stored; needs a supply follow-up task.
vocabulary.released.v1	records	intake, crm-engine, listings, insight	A new controlled-vocabulary release is active.
photo.added.v1	records	listings, insight	Photo attached to a property.
photo.removed.v1	records	listings, insight	Photo detached or deleted.

Event	Producer	Consumers	Meaning
project.created.v1	records	listings, insight	A Project (Sale, Primary) exists.
project.updated.v1	records	listings, insight	Project facts changed (full current facts).
price_sheet.applied.v1	records	journeys, crm-engine, insight	A developer price sheet was applied to a project (life-curve basis for Sale, Primary).
offer.voided.v1	records	journeys, crm-engine, listings, insight	An offer was voided because review changed its side or scope (it was not supply).
demand.voided.v1	records	journeys, crm-engine, listings, insight	A demand was voided because review changed its side or scope.
person.flag_removed.v1	records	journeys, insight	A person flag was removed.
desk_item.created.v1	records	insight	A Business, Capital, Equipment or Market Participant record was stored.
desk_item.updated.v1	records	insight	A desk item changed (assigned, archived).
merge_candidate.raised.v1	records	insight	An uncertain merge or price gap needs review.
market_data.recorded.v1	records	insight	A closed or reported price was kept as market data (ours, elsewhere, or reported).
micromarkets.updated.v1	records	crm-engine, listings, insight	The micromarket hierarchy or adjacency changed.
offer.confirmed.v1	journeys	records, crm-engine, listings, insight	Offer confirmed on a call/visit/price sheet; life curve reset.
demand.confirmed.v1	journeys	crm-engine, insight	Demand reconfirmed; life curve reset.
lifecycle.stage_changed.v1	journeys	crm-engine, listings, insight	Life curve stage changed.
offer.commercial_status_changed.v1	journeys	crm-engine, listings, insight	Offer Commercial axis changed.
demand.qualified.v1	journeys	crm-engine, insight	Demand qualified; run inventory check.
demand.status_changed.v1	journeys	crm-engine, listings, insight	Demand Commercial axis changed.
demand.exited.v1	journeys	records, crm-engine, listings, insight	Demand exited Lost/Dormant/Invalid.
demand.reactivated.v1	journeys	records, crm-engine, insight	Dormant demand returned to Active.
demand.sourcing_started.v1	journeys	listings, insight	Demand in Sourcing; anonymous demand post allowed.
sourcing_request.created.v1	journeys	insight	Sourcing request raised.
proposal.sent.v1	journeys	crm-engine, insight	Proposal marked sent.
site_visit.completed.v1	journeys	records, crm-engine, insight	Visit done; resets both life curves.
deal.opened.v1	journeys	crm-engine, insight	Deal entered In process.
deal.closed.v1	journeys	records, crm-engine, listings, insight	Deal closed; offer closed.
deal.cancelled.v1	journeys	records, crm-engine, listings, insight	Deal cancelled; offer back to Available, demand to Active (compensation).

Event	Producer	Consumers	Meaning
offer.retired.v1	journeys	records, crm-engine, listings, insight	Offer Inactive (already gone / unwilling).
lease_renewal.due.v1	journeys	records	Month 10 of an 11-month lease closed by 11 Estates: create an Upcoming offer.
watchlist_task.completed.v1	journeys	insight	Supply follow-up on a Market Signal done.
call.logged.v1	journeys	records, insight	A call outcome was recorded.
site_visit.scheduled.v1	journeys	insight	A site visit was scheduled.
sourcing_request.updated.v1	journeys	insight	Sourcing request status changed.
proposal.feedback_recorded.v1	journeys	crm-engine, insight	Client feedback on proposal options (feeds match usefulness M6).
deal.updated.v1	journeys	insight	Deal stage or follow-up changed.
queue.counts_changed.v1	journeys	insight	Per-user queue section counts changed (debounced $\leq 1/\text{min}$ per user).
match.suggested.v1	crm-engine	journeys, insight	New or re-ranked suggestion.
match.confirmed.v1	crm-engine	journeys, listings, insight	A person confirmed a match.
match.rejected.v1	crm-engine	journeys, insight	A person rejected a match.
match.closed.v1	crm-engine	journeys, insight	Match closed automatically (offer closed/retired, demand exited).
match.flagged.v1	crm-engine	journeys, insight	Match flag raised (price above budget, reconfirm, area/market unknown).
demand.matching_completed.v1	crm-engine	journeys, insight	Inventory check for a demand finished (after qualify or re-run).
match.reopened.v1	crm-engine	journeys, insight	A closed match reopened (compensation, e.g. deal cancelled or merge undone).
publication.changed.v1	listings	records, journeys, insight	Publication level of an offer/project/demand post changed.
export.completed.v1	insight	web	Export file ready (24 h link).
export.failed.v1	insight	web	Export could not be produced.
user.changed.v1	web	journeys, insight	A staff user was invited, changed role, or deactivated.
audit.recorded.v1	*	web	Sensitive action for the audit log (contact view, export, merge/undo, publication, exit, deal close, settings). Automatic actions use the reserved system user 00000000-0000-0000-0000-000000000001.

Endpoint catalogue (generated from contracts/openapi)

Auth: staff = staffViaWeb (signed-in user via web), svc = serviceToken, cron = X-Cron-Secret, key = website API key, client = X-Client-Secret, token = proposal share token, none = public health.

Summary

Service	Operations	Public (API key)	Internal (svc/cron)
web	17	0	4
intake	30	0	5
records	69	0	6
journeys	60	0	4
crm-engine	19	0	4
listings	25	6	3
insight	20	0	3
Total	240		

web (17)

Method	Path	Auth	Roles	Idempotency	Paged	Rate limit	Timeout ms	Emits	Summary
GET	/v1/me	staffSession	all	safe		api bucket (20 req/s, burst 40 per user)	2000	—	Signed-in user, role, tenant, permissions, session expiry and environment flags (drives the UI)
DELETE	/v1/me/session	staffSession	all	by design		api bucket	2000	—	Sign out (revokes the Supabase refresh token, clears cookies, drops cached service tokens)
GET	/v1/me/notifications	staffSession	all	safe	yes	api bucket	2000	—	The caller's upload, export and user notifications (web-owned, R-6); journeys owns /v1/notifications for work notifications
POST	/v1/me/notifications/read	staffSession	all	Idempotency-Key		api bucket	2000	—	Mark notifications read (listed ids, or everything up to a time)
GET	/v1/roles	staffSession	all	safe		api bucket	2000	—	The five staff roles and their permission codes (PRD §2.3), read-only
GET	/v1/users	staffSession	all	safe	yes	api bucket	2000	—	Users of the tenant (Settings → Users & roles). Non-admins get a directory view (id, name, role, status) for pickers.
POST	/v1/users/invitations	staffSession	Admin	Idempotency-Key		api bucket; plus 20 invitations per hour per tenant	2000	user.changed.v1	Invite a Google account by email with a role (Supabase Auth invite; expires in 7 days)
DELETE	/v1/users/invitations/{userId}	staffSession	Admin	by design		api bucket	2000	user.changed.v1	Revoke a pending invitation (the user row is marked deactivated; the Supabase invite is deleted)

Method	Path	Auth	Roles	Idempotency	Paged	Rate limit	Timeout ms	Emits	Summary
PATCH	/v1/users/{userId}	staffSession	Admin	If-Match		api bucket	2000	user.changed.v1	Change role, data-operator flag, display name, or deactivate / reactivate (JSON Merge Patch)
GET	/v1/audit-log	staffSession	Admin	safe	yes	api bucket	2000	—	Immutable audit log (US-35), newest first; filters on stored fields
GET	/ .well-known/jwks.json	none	service	safe		none (CDN cached, max-age=600)	500	—	Public keys that verify web-minted service tokens (current + previous during rotation)
POST	/internal/v1/service-tokens	serviceCredential	service	by design		60 per minute per caller service	2000	—	Mint a service-to-service token (aud=, sub=, tid, 5 min) for projection rebuilds and background jobs
POST	/internal/v1/relay	cron	scheduler	by design		none (scheduler only; single-flight via advisory lock)	55000	user.changed.v1	Publish unpublished outbox rows (user.changed.v1) to consumer queues (pg_cron every minute + poke after commit)
POST	/internal/v1/drain/{queue}	cron	scheduler	by design		none (scheduler only; every minute + poke from producer relays)	55000	—	Consume q_web: audit.recorded.v1 → audit_log; upload.completed.v1 / upload.failed.v1 / export.completed.v1 / export.failed.v1 → notifications
POST	/internal/v1/jobs/{name}	cron	scheduler	by design		none (scheduler only; single-flight per job)	55000	—	Scheduled jobs
GET	/health/live	none	anonymous	safe		none	500	—	
GET	/health/ready	none	anonymous	safe		none	1000	—	

intake (30)

Method	Path	Auth	Roles	Idempotency	Paged	Rate limit	Timeout ms	Emits	Summary
POST	/v1/uploads	staff	all	Idempotency-Key		5 uploads/hour per user (conventions §4), plus 20 req/s per user on reads	2000	—	Create an upload and get a signed storage URL
GET	/v1/uploads	staff	all	safe	yes	20 req/s per user, burst 40 (enforced at web, conventions §4)	2000	—	List uploads (newest first)
GET	/v1/uploads/{idOrCode}	staff	all	safe		20 req/s per user, burst 40 (enforced at web, conventions §4)	2000	—	Get an upload (status, mode, counts, suggested mapping)
PATCH	/v1/uploads/{idOrCode}	staff	all	If-Match		20 req/s per user, burst 40 (enforced at web, conventions §4)	2000	—	Change pre-start options (anonymise switch, source detail, sheet)
POST	/v1/uploads/{idOrCode}/inspect	staff	all	Idempotency-Key		20 req/s per user, burst 40 (enforced at web, conventions §4)	2000	—	Confirm the file is in storage and inspect it asynchronously

Method	Path	Auth	Roles	Idempotency	Paged	Rate limit	Timeout ms	Emits	Summary
PUT	/v1/uploads/{idOrCode}/mapping	staff	all	by design		20 req/s per user, burst 40 (enforced at web, conventions §4)	2000	—	Set the column mapping (mapping mode) and optionally save it as a template
POST	/v1/uploads/{idOrCode}/start	staff	all	Idempotency-Key		20 req/s per user, burst 40 (enforced at web, conventions §4)	2000	upload.started.v1, rows.classified.v1, review_item.created.v1, upload.completed.v1, upload.failed.v1	Start background processing
POST	/v1/uploads/{idOrCode}/cancel	staff	all	Idempotency-Key		20 req/s per user, burst 40 (enforced at web, conventions §4)	2000	upload.failed.v1	Cancel an upload that has not completed
GET	/v1/uploads/{idOrCode}/progress	staff	all	safe		20 req/s per user, burst 40 (enforced at web, conventions §4)	2000	—	Processing progress per stage
GET	/v1/uploads/{idOrCode}/row-errors	staff	all	safe	yes	20 req/s per user, burst 40 (enforced at web, conventions §4)	2000	—	Row errors (rejected rows) with field, code and non-PII value
GET	/v1/uploads/{idOrCode}/rejected-rows	staff	all	safe		20 req/s per user, burst 40 (enforced at web, conventions §4)	2000	audit.recorded.v1	Signed link to download the rejected rows file (CSV, original values)
GET	/v1/uploads/{idOrCode}/migration-map	staff	all	safe	yes	20 req/s per user, burst 40 (enforced at web, conventions §4)	2000	—	Parsed migration_map sheet (CR-006 Z-5)
GET	/v1/templates	staff	all	safe	yes	20 req/s per user, burst 40 (enforced at web, conventions §4)	2000	—	List saved mapping templates
POST	/v1/templates	staff	all	Idempotency-Key		20 req/s per user, burst 40 (enforced at web, conventions §4)	2000	—	Create a mapping template
GET	/v1/templates/{id}	staff	all	safe		20 req/s per user, burst 40 (enforced at web, conventions §4)	2000	—	Get a template
PUT	/v1/templates/{id}	staff	all	by design		20 req/s per user, burst 40 (enforced at web, conventions §4)	2000	—	Replace a template
DELETE	/v1/templates/{id}	staff	Admin, Manager, Data operator	by design		20 req/s per user, burst 40 (enforced at web, conventions §4)	2000	—	Delete (soft) a template
GET	/v1/review-items	staff	Admin, Manager, Data operator	safe	yes	20 req/s per user, burst 40 (enforced at web, conventions §4)	2000	—	Classification review queue, filterable by reason code

Method	Path	Auth	Roles	Idempotency	Paged	Rate limit	Timeout ms	Emits	Summary
GET	/v1/review-items/summary	staff	Admin, Manager, Data operator	safe		20 req/s per user, burst 40 (enforced at web, conventions §4)	2000	—	Open review items grouped by reason code (C-05 group headers)
GET	/v1/review-items/{id}	staff	Admin, Manager, Data operator	safe		20 req/s per user, burst 40 (enforced at web, conventions §4)	2000	—	Get one review item with redacted context
POST	/v1/review-items/{id}/resolve	staff	Admin, Manager, Data operator	Idempotency-Key		20 req/s per user, burst 40 (enforced at web, conventions §4)	2000	review_item.resolved.v1	Resolve a review item (set / confirm / skip)
POST	/v1/review-items/bulk-resolve	staff	Admin, Manager, Data operator	Idempotency-Key		20 req/s per user, burst 40 (enforced at web, conventions §4)	2000	review_item.resolved.v1	Resolve up to 100 review items with one classification (e.g. accept defaulted side)
POST	/v1/parse	staff	Admin, Manager, Demand agent, Supply agent	by design		30 req/min per user (bounds model spend; conventions §4 default otherwise)	4000	—	Parse typed free text into a suggested classification (quick add prefill)
GET	/internal/v1/uploads/{uploadId}/rows	svc	—	safe		records drainers only (≤ 8 concurrent); no user limit	2000	—	Rows of one emitted batch, with PII and raw text (records ingestion)
GET	/internal/v1/uploads/{uploadId}/migration-map	svc	—	safe	yes	records drainers only; no user limit	2000	—	Migration map entries for records to apply before any batch of this upload
POST	/internal/v1/relay	cron	—	by design		n/a: pg_cron/pg_net only (1 call/min per job plus pokes)	60000	—	Relay unpublished outbox rows to consumer pgmq queues
POST	/internal/v1/drain/{queue}	cron	—	by design		n/a: pg_cron/pg_net only (1 call/min per job plus pokes)	60000	—	Drain one of this service's pgmq queues
POST	/internal/v1/jobs/{name}	cron	—	by design		n/a: pg_cron/pg_net only (1 call/min per job plus pokes)	60000	—	Run a scheduled job
GET	/health/live	none	—	safe		none (health probe)	2000	—	Liveness (process up)
GET	/health/ready	none	—	safe		none (health probe)	2000	—	Readiness (DB reachable, migrations at expected version)

records (69)

Method	Path	Auth	Roles	Idempotency	Paged	Rate limit	Timeout ms	Emits	Summary
GET	/v1/offers	svc, staff	all	safe	yes	20 req/s per user, burst 40 (enforced at web, conventions §4)	2000	—	Search offers by stored fields

Method	Path	Auth	Roles	Idempotency	Paged	Rate limit	Timeout ms	Emits	Summary
POST	/v1/offers	staff	Admin, Manager, Supply agent	Idempotency-Key		20 req/s per user, burst 40 (enforced at web, conventions §4)	2000	offer.created.v1	Add an offer to an existing property (US-13)
GET	/v1/offers/{idOrCode}	svc, staff	all	safe		20 req/s per user, burst 40 (enforced at web, conventions §4)	2000	—	Get an offer
PATCH	/v1/offers/{idOrCode}	staff	Admin, Manager, Supply agent, Demand agent	If-Match		20 req/s per user, burst 40 (enforced at web, conventions §4)	2000	offer.updated.v1, offer.price_changed.v1	Update offer facts (prices, deal tags, units, revenue share, owner)
POST	/v1/offers/{idOrCode}/record-stage	staff	Admin, Manager, Supply agent, Demand agent	Idempotency-Key		20 req/s per user, burst 40 (enforced at web, conventions §4)	2000	offer.record_stage_changed.v1, offer.updated.v1	Move the offer's Record axis
PUT	/v1/offers/{idOrCode}/photos	staff	Admin, Manager, Supply agent, Demand agent	by design		20 req/s per user, burst 40 (enforced at web, conventions §4)	2000	offer.updated.v1	Choose which property photos this offer uses (D-5)
GET	/v1/properties	svc, staff	all	safe	yes	20 req/s per user, burst 40 (enforced at web, conventions §4)	2000	—	Search properties by stored fields
POST	/v1/properties	staff	Admin, Manager, Supply agent	Idempotency-Key		20 req/s per user, burst 40 (enforced at web, conventions §4)	2000	offer.created.v1	Create a property with 1–4 offers (manual supply entry)
POST	/v1/properties/dedup-check	staff	Admin, Manager, Supply agent, Demand agent	by design		20 req/s per user, burst 40 (enforced at web, conventions §4)	2000	—	Property-level duplicate check without creating anything
GET	/v1/properties/{idOrCode}	svc, staff	all	safe		20 req/s per user, burst 40 (enforced at web, conventions §4)	2000	—	Get a property with its offers and parties
PATCH	/v1/properties/{idOrCode}	staff	Admin, Manager, Supply agent, Demand agent	If-Match		20 req/s per user, burst 40 (enforced at web, conventions §4)	2000	offer.updated.v1	Update property facts (areas, location, building, unit details)
GET	/v1/properties/{idOrCode}/sightings	staff	all	safe	yes	20 req/s per user, burst 40 (enforced at web, conventions §4)	2000	—	Sightings (raw appearances) of the property and its offers
GET	/v1/properties/{idOrCode}/second-sources	staff	all	safe	yes	20 req/s per user, burst 40 (enforced at web, conventions §4)	2000	—	Other sources advertising the same property, with price gaps
GET	/v1/properties/{idOrCode}/photos	svc, staff	all	safe	yes	20 req/s per user, burst 40 (enforced at web, conventions §4)	2000	—	Photos of a property (signed read URLs)

Method	Path	Auth	Roles	Idempotency	Paged	Rate limit	Timeout ms	Emits	Summary
GET	/v1/second-sources	staff	Admin, Manager, Data operator, Supply agent	safe	yes	20 req/s per user, burst 40 (enforced at web, conventions §4)	2000	—	Price-gap review queue (open second sources with a > 5% gap)
POST	/v1/second-sources/{id}/resolve	staff	Admin, Manager, Data operator, Supply agent	Idempotency-Key		20 req/s per user, burst 40 (enforced at web, conventions §4)	2000	offer.price_changed.v1, offer.updated.v1	Accept the second source's price or dismiss the gap
GET	/v1/projects	svc, staff	all	safe	yes	20 req/s per user, burst 40 (enforced at web, conventions §4)	2000	—	Search projects
POST	/v1/projects	staff	Admin, Manager, Supply agent	Idempotency-Key		20 req/s per user, burst 40 (enforced at web, conventions §4)	2000	project.created.v1	Create a project (Sale, Primary)
GET	/v1/projects/{idOrCode}	svc, staff	all	safe		20 req/s per user, burst 40 (enforced at web, conventions §4)	2000	—	Get a project with its configuration offers
PATCH	/v1/projects/{idOrCode}	staff	Admin, Manager, Supply agent	If-Match		20 req/s per user, burst 40 (enforced at web, conventions §4)	2000	project.updated.v1, offer.updated.v1	Update project facts (RERA, possession, amenities)
POST	/v1/projects/{idOrCode}/price-sheets	staff	Admin, Manager, Supply agent	Idempotency-Key		20 req/s per user, burst 40 (enforced at web, conventions §4)	2000	price_sheet.applied.v1, project.updated.v1, offer.created.v1, offer.updated.v1, offer.price_changed.v1	Apply a developer price sheet: create/update configuration offers, prices and units (US-17)
GET	/v1/projects/{idOrCode}/price-sheets	staff	all	safe	yes	20 req/s per user, burst 40 (enforced at web, conventions §4)	2000	—	Price sheet history (newest first)
GET	/v1/demands	svc, staff	all	safe	yes	20 req/s per user, burst 40 (enforced at web, conventions §4)	2000	—	Search demands by stored fields
POST	/v1/demands	staff	Admin, Manager, Demand agent	Idempotency-Key		20 req/s per user, burst 40 (enforced at web, conventions §4)	2000	demand.created.v1, demand.touch_added.v1	Create a demand (full form; quick add uses /v1/quick-add)
GET	/v1/demands/{idOrCode}	svc, staff	all	safe		20 req/s per user, burst 40 (enforced at web, conventions §4)	2000	—	Get a demand
PATCH	/v1/demands/{idOrCode}	staff	Admin, Manager, Demand agent	If-Match		20 req/s per user, burst 40 (enforced at web, conventions §4)	2000	demand.updated.v1	Update demand facts
POST	/v1/demands/{idOrCode}/record-stage	staff	Admin, Manager, Demand agent	Idempotency-Key		20 req/s per user, burst 40 (enforced at web, conventions §4)	2000	demand.updated.v1	Move the demand's Record axis

Method	Path	Auth	Roles	Idempotency	Paged	Rate limit	Timeout ms	Emits	Summary
GET	/v1/demands/{idOrCode}/touches	staff	all	safe	yes	20 req/s per user, burst 40 (enforced at web, conventions §4)	2000	—	Touches of a demand (first touch flagged)
POST	/v1/demands/{idOrCode}/touches	staff	Admin, Manager, Demand agent	Idempotency-Key		20 req/s per user, burst 40 (enforced at web, conventions §4)	2000	demand.touch_added.v1	Record another arrival of the same demand
POST	/v1/demands/{idOrCode}/add-supply	staff	Admin, Manager, Supply agent, Demand agent	Idempotency-Key		20 req/s per user, burst 40 (enforced at web, conventions §4)	2000	offer.created.v1, offer.record_stage_changed.v1	Add supply found for a demand or SRQ (US-05, C-07)
GET	/v1/people	svc, staff	all	safe	yes	20 req/s per user, burst 40 (enforced at web, conventions §4)	2000	—	Search people (no contact fields)
POST	/v1/people	staff	Admin, Manager, Supply agent, Demand agent	Idempotency-Key		20 req/s per user, burst 40 (enforced at web, conventions §4)	2000	—	Create a person (phones deduplicated)
GET	/v1/people/{idOrCode}	svc, staff	all	safe		20 req/s per user, burst 40 (enforced at web, conventions §4)	2000	—	Get a person (masked)
PATCH	/v1/people/{idOrCode}	staff	Admin, Manager, Supply agent, Demand agent	If-Match		20 req/s per user, burst 40 (enforced at web, conventions §4)	2000	—	Update a person (contacts write-only)
POST	/v1/people/{idOrCode}/flags	staff	Admin, Manager, Supply agent, Demand agent	Idempotency-Key		20 req/s per user, burst 40 (enforced at web, conventions §4)	2000	person.flagged.v1, person.flag_removed.v1	Add or remove a person flag
POST	/v1/quick-add/lookup	staff	Admin, Manager, Demand agent, Supply agent	by design		20 req/s per user, burst 40 (enforced at web, conventions §4)	2000	—	Phone-first lookup: matching people, their open demands, offers and flags (US-04 AC1)
POST	/v1/quick-add	staff	Admin, Manager, Demand agent, Supply agent	Idempotency-Key		20 req/s per user, burst 40 (enforced at web, conventions §4)	2000	demand.created.v1, demand.touch_added.v1, offer.created.v1, merge_candidate.raised.v1	Quick add: create a demand or supply, or add a touch to an existing demand
POST	/v1/reveals	staff	all	Idempotency-Key		60 reveals/hour per user (anti-scraping; justification: normal call work is < 40 contacts/day, D-10)	2000	audit.recorded.v1	Reveal contact / unit-level fields of one subject (audit-logged, R-VIS-3)
GET	/v1/enquiries	svc, staff	all	safe	yes	20 req/s per user, burst 40 (enforced at web, conventions §4)	2000	—	List enquiries
GET	/v1/enquiries/{idOrCode}	svc, staff	all	safe		20 req/s per user, burst 40 (enforced at web, conventions §4)	2000	—	Get an enquiry (message via reveal)

Method	Path	Auth	Roles	Idempotency	Paged	Rate limit	Timeout ms	Emits	Summary
GET	/v1/source-ads	staff	all	safe	yes	20 req/s per user, burst 40 (enforced at web, conventions §4)	2000	—	List source ads (one uploaded ad → 1..n records)
GET	/v1/source-ads/{idOrCode}	staff	all	safe		20 req/s per user, burst 40 (enforced at web, conventions §4)	2000	—	Get a source ad with its split children
GET	/v1/merge-candidates	staff	Admin, Manager, Data operator	safe	yes	20 req/s per user, burst 40 (enforced at web, conventions §4)	2000	—	Uncertain-merge review queue
POST	/v1/merge-candidates/{id}/dismiss	staff	Admin, Manager, Data operator	Idempotency-Key		20 req/s per user, burst 40 (enforced at web, conventions §4)	2000	—	Mark a candidate Different or Skip
POST	/v1/merges	staff	Admin, Manager, Data operator	Idempotency-Key		20 req/s per user, burst 40 (enforced at web, conventions §4)	2000	records.merged.v1, audit.recorded.v1, demand.touch_added.v1, offer.updated.v1, demand.updated.v1	Merge records into a survivor (reversible)
GET	/v1/merges/{id}	staff	all	safe		20 req/s per user, burst 40 (enforced at web, conventions §4)	2000	—	Get a merge and what it moved
POST	/v1/merges/{id}/undo	staff	Admin, Manager	Idempotency-Key		20 req/s per user, burst 40 (enforced at web, conventions §4)	2000	records.merge_undone.v1, audit.recorded.v1, offer.updated.v1, demand.updated.v1	Undo a merge (US-09)
GET	/v1/desks/{desk}	staff	all	safe	yes	20 req/s per user, burst 40 (enforced at web, conventions §4)	2000	—	Desk list view (US-36)
GET	/v1/desk-items/{idOrCode}	staff	all	safe		20 req/s per user, burst 40 (enforced at web, conventions §4)	2000	—	Get a desk item
PATCH	/v1/desk-items/{idOrCode}	staff	Admin, Manager	If-Match		20 req/s per user, burst 40 (enforced at web, conventions §4)	2000	desk_item.updated.v1	Assign, archive or annotate a Business / Capital / Archive / Watchlist item
GET	/v1/vocabulary	svc, staff	all	safe		20 req/s per user, burst 40 (enforced at web, conventions §4)	2000	—	The active controlled-vocabulary release (read-only, D-16)
GET	/v1/vocabulary/versions	svc, staff	all	safe	yes	20 req/s per user, burst 40 (enforced at web, conventions §4)	2000	—	Vocabulary release history
GET	/v1/micromarkets	svc, staff	all	safe	yes	20 req/s per user, burst 40 (enforced at web, conventions §4)	2000	—	Micromarket hierarchy (zone → micromarket → locality → sub-locality, with aliases)

Method	Path	Auth	Roles	Idempotency	Paged	Rate limit	Timeout ms	Emits	Summary
POST	/v1/micromarkets	staff	Admin	Idempotency-Key		20 req/s per user, burst 40 (enforced at web, conventions §4)	2000	micromarkets.updated.v1	Add a hierarchy node (Admin, US-34)
PATCH	/v1/micromarkets/{id}	staff	Admin	If-Match		20 req/s per user, burst 40 (enforced at web, conventions §4)	2000	micromarkets.updated.v1	Rename, re-parent, change aliases or adjacency (Admin, R-13)
GET	/v1/launch-area	svc, staff	all	safe		20 req/s per user, burst 40 (enforced at web, conventions §4)	2000	—	Cities enabled for queues, matching and listings (CR-006 Z-7)
PUT	/v1/launch-area	staff	Admin	by design		20 req/s per user, burst 40 (enforced at web, conventions §4)	2000	offer.updated.v1, demand.updated.v1	Replace the enabled city list (Admin)
POST	/v1/photos	staff	Admin, Manager, Supply agent, Demand agent	Idempotency-Key		20 req/s per user, burst 40 (enforced at web, conventions §4)	2000	—	Get a signed URL to upload a photo for a property
POST	/v1/photos/{id}/attach	staff	Admin, Manager, Supply agent, Demand agent	Idempotency-Key		20 req/s per user, burst 40 (enforced at web, conventions §4)	2000	photo.added.v1, offer.updated.v1	Confirm the upload and attach the photo to the property (and chosen offers)
DELETE	/v1/photos/{id}	staff	Admin, Manager, Supply agent, Demand agent	by design		20 req/s per user, burst 40 (enforced at web, conventions §4)	2000	photo.removed.v1, offer.updated.v1	Detach and delete a photo
GET	/internal/v1/properties/{id}/scan-terms	svc	—	safe		listings only; no user limit	2000	—	Salted hashes of building/society name tokens, wing and unit for the privacy scan (R-20)
GET	/internal/v1/photos/{id}/signed-url	svc	—	safe		listings only; no user limit	2000	—	Short-lived signed URL to read an original photo so listings can build public renditions (Stage 4 reconciliation L-2)
POST	/internal/v1/contacts:batch	svc	—	by design		insight export workers only; no user limit	2000	audit.recorded.v1	Contacts of up to 1,000 people for an export (R-21, audited)
GET	/v1/market-data	svc, staff	all	safe	yes	20 req/s per user, burst 40 (enforced at web, conventions §4)	2000	—	Market data points (closed / reported prices)
POST	/internal/v1/relay	cron	—	by design		n/a: pg_cron/pg_net only (1 call/min per job plus pokes)	60000	—	Relay unpublished outbox rows to consumer pgmq queues
POST	/internal/v1/drain/{queue}	cron	—	by design		n/a: pg_cron/pg_net only (1 call/min per job plus pokes)	60000	—	Drain one of this service's pgmq queues
POST	/internal/v1/jobs/{name}	cron	—	by design		n/a: pg_cron/pg_net only (1 call/min per job plus pokes)	60000	—	Run a scheduled job
GET	/health/live	none	—	safe		none (health probe)	2000	—	Liveness (process up)

Method	Path	Auth	Roles	Idempotency	Paged	Rate limit	Timeout ms	Emits	Summary
GET	/health/ready	none	—	safe		none (health probe)	2000	—	Readiness (DB reachable, migrations at expected version)

journeys (60)

Method	Path	Auth	Roles	Idempotency	Paged	Rate limit	Timeout ms	Emits	Summary
GET	/v1/queues/me	staff	Admin, Manager, Demand agent, Supply agent	safe		20/s per user, burst 40 (token bucket at web)	2000	—	My queue summary: sections with counts, capacity and today's plan (P-01, C-01)
GET	/v1/queues/me/sections/{section}	staff	Admin, Manager, Demand agent, Supply agent	safe	yes	20/s per user, burst 40 (token bucket at web)	2000	—	Items of one section of my queue, in computed order
GET	/v1/queues/users/{userId}	staff	Admin, Manager	safe		20/s per user, burst 40 (token bucket at web)	2000	—	A team member's queue summary (Manager view)
GET	/v1/queues/users/{userId}/sections/{section}	staff	Admin, Manager	safe	yes	20/s per user, burst 40 (token bucket at web)	2000	—	Items of one section of a team member's queue (Manager view)
POST	/v1/queue-items/reassign	staff	Admin, Manager	Idempotency-Key		20/s per user, burst 40 (token bucket at web)	2000	audit.recorded.v1	Bulk reassign open queue items (≤ 100) to another user (C-19)
GET	/v1/capacities	staff	Admin, Manager	safe	yes	20/s per user, burst 40 (token bucket at web)	2000	—	Daily call capacities of all users (D-10)
GET	/v1/capacities/{userId}	staff	Admin, Manager, Demand agent, Supply agent	safe		20/s per user, burst 40 (token bucket at web)	2000	—	One user's daily capacity (self, or Manager/Admin for anyone)
PUT	/v1/capacities/{userId}	staff	Admin, Manager	by design		20/s per user, burst 40 (token bucket at web)	2000	audit.recorded.v1	Set a user's daily call capacity (default 40, A-36)
POST	/v1/calls	staff	Admin, Manager, Demand agent, Supply agent	Idempotency-Key		20/s per user, burst 40 (token bucket at web)	2000	call.logged.v1, offer.confirmed.v1, demand.confirmed.v1, lifecycle.stage_changed.v1, offer.commercial_status_changed.v1, demand.status_changed.v1, demand.reactivated.v1, offer.retired.v1	Log a call or meeting outcome (C-08); drives life curve, attempts, retire
GET	/v1/calls	staff	all	safe	yes	20/s per user, burst 40 (token bucket at web)	2000	—	Call history for a subject or a person (P-04)
GET	/v1/life-curves/{subjectType}/{subjectId}	staff	all	safe		20/s per user, burst 40 (token bucket at web)	2000	—	Life curve for an offer or demand (stage, day count, thresholds)

Method	Path	Auth	Roles	Idempotency	Paged	Rate limit	Timeout ms	Emits	Summary
GET	/v1/offers/{idOrCode}/journey	staff	all	safe		20/s per user, burst 40 (token bucket at web)	2000	—	Offer work state: Commercial axis, life curve, queue, signals (P-02)
POST	/v1/offers/{idOrCode}/retire	staff	Admin, Manager, Supply agent	Idempotency-Key		20/s per user, burst 40 (token bucket at web)	2000	offer.retired.v1, offer.commercial_status_changed.v1, audit.recorded.v1	Retire an offer to Inactive (already gone / unwilling / other) (C-17, US-18)
GET	/v1/demands/{idOrCode}/journey	staff	all	safe		20/s per user, burst 40 (token bucket at web)	2000	—	Demand work state: Commercial axis, exit, qualification, life curve (P-03)
POST	/v1/demands/{idOrCode}/qualify	staff	Admin, Manager, Demand agent	Idempotency-Key		20/s per user, burst 40 (token bucket at web)	2000	demand.qualified.v1, demand.status_changed.v1, demand.confirmed.v1, lifecycle.stage_changed.v1	Mark a demand qualified (C-09); triggers the inventory check
POST	/v1/demands/{idOrCode}/exit	staff	Admin, Manager, Demand agent	Idempotency-Key		20/s per user, burst 40 (token bucket at web)	2000	demand.exited.v1, lifecycle.stage_changed.v1, sourcing_request.updated.v1, audit.recorded.v1	Exit a demand: Lost, Dormant or Invalid (C-16, US-26)
POST	/v1/demands/{idOrCode}/reactivate	staff	Admin, Manager, Demand agent	Idempotency-Key		20/s per user, burst 40 (token bucket at web)	2000	demand.reactivated.v1, demand.status_changed.v1, lifecycle.stage_changed.v1, demand.confirmed.v1, audit.recorded.v1	Return an exited demand to Active (Dormant revisit; Lost/Invalid = Manager override)
POST	/v1/sourcing-requests	staff	Admin, Manager, Demand agent	Idempotency-Key		20/s per user, burst 40 (token bucket at web)	2000	sourcing_request.created.v1, demand.sourcing_started.v1, demand.status_changed.v1	Raise a sourcing request to the supply team (C-11, US-22)
GET	/v1/sourcing-requests	staff	all	safe	yes	20/s per user, burst 40 (token bucket at web)	2000	—	List sourcing requests
GET	/v1/sourcing-requests/{idOrCode}	staff	all	safe		20/s per user, burst 40 (token bucket at web)	2000	—	Get a sourcing request
PATCH	/v1/sourcing-requests/{idOrCode}	staff	Admin, Manager, Demand agent, Supply agent	If-Match		20/s per user, burst 40 (token bucket at web)	2000	sourcing_request.updated.v1, demand.status_changed.v1	Update status, assignee, due date or priority
POST	/v1/proposals	staff	Admin, Manager, Demand agent	Idempotency-Key		20/s per user, burst 40 (token bucket at web)	2000	—	Create a proposal from confirmed matches (C-13, US-23)
GET	/v1/proposals	staff	all	safe	yes	20/s per user, burst 40 (token bucket at web)	2000	—	List proposals
GET	/v1/proposals/{idOrCode}	staff	all	safe		20/s per user, burst 40 (token bucket at web)	2000	—	Get a proposal

Method	Path	Auth	Roles	Idempotency	Paged	Rate limit	Timeout ms	Emits	Summary
PATCH	/v1/proposals/{idOrCode}	staff	Admin, Manager, Demand agent	If-Match		20/s per user, burst 40 (token bucket at web)	2000	—	Change options or cover note (not after Sent)
POST	/v1/proposals/{idOrCode}/pdf	staff	Admin, Manager, Demand agent	Idempotency-Key		20/s per user, burst 40 (token bucket at web)	2000	—	Queue PDF generation (async, off the request path)
GET	/v1/proposals/{idOrCode}/pdf	staff	Admin, Manager, Demand agent	safe		20/s per user, burst 40 (token bucket at web)	2000	—	PDF status and a 5-minute signed download URL
POST	/v1/proposals/{idOrCode}/share-link	staff	Admin, Manager, Demand agent	Idempotency-Key		20/s per user, burst 40 (token bucket at web)	2000	audit.recorded.v1	Create (or replace) the private share link, 14-day expiry (A-40)
DELETE	/v1/proposals/{idOrCode}/share-link	staff	Admin, Manager, Demand agent	by design		20/s per user, burst 40 (token bucket at web)	2000	audit.recorded.v1	Revoke the active share link
POST	/v1/proposals/{idOrCode}/mark-sent	staff	Admin, Manager, Demand agent	Idempotency-Key		20/s per user, burst 40 (token bucket at web)	2000	proposal.sent.v1, offer.commercial_status_changed.v1, demand.status_changed.v1	Mark the proposal sent (logs date and channel; the system sends nothing)
POST	/v1/proposals/{idOrCode}/feedback	staff	Admin, Manager, Demand agent	Idempotency-Key		20/s per user, burst 40 (token bucket at web)	2000	proposal.feedback_recorded.v1	Record client feedback per option
GET	/p/{token}	none	—	safe		30/min per IP and 120/min per token (enforced at web edge)	2000	—	Public proposal page for a share-link token (no staff auth)
POST	/v1/site-visits	staff	Admin, Manager, Demand agent, Supply agent	Idempotency-Key		20/s per user, burst 40 (token bucket at web)	2000	site_visit.scheduled.v1	Schedule a site visit (C-14)
GET	/v1/site-visits	staff	all	safe	yes	20/s per user, burst 40 (token bucket at web)	2000	—	List site visits
GET	/v1/site-visits/{idOrCode}	staff	all	safe		20/s per user, burst 40 (token bucket at web)	2000	—	Get a site visit
PATCH	/v1/site-visits/{idOrCode}	staff	Admin, Manager, Demand agent, Supply agent	If-Match		20/s per user, burst 40 (token bucket at web)	2000	—	Reschedule, change offers/attendees or cancel
POST	/v1/site-visits/{idOrCode}/complete	staff	Admin, Manager, Demand agent, Supply agent	Idempotency-Key		20/s per user, burst 40 (token bucket at web)	2000	site_visit.completed.v1, offer.confirmed.v1, demand.confirmed.v1, lifecycle.stage_changed.v1, offer.commercial_status_changed.v1, demand.status_changed.v1	Record the visit outcome; resets both life curves (US-24)

Method	Path	Auth	Roles	Idempotency	Paged	Rate limit	Timeout ms	Emits	Summary
POST	/v1/deals	staff	Admin, Manager, Demand agent	Idempotency-Key		20/s per user, burst 40 (token bucket at web)	2000	deal.opened.v1, deal.updated.v1, offer.commercial_status_changed.v1, demand.status_changed.v1	Open a deal (In process) for one demand and one offer (C-15, US-25)
GET	/v1/deals	staff	all	safe	yes	20/s per user, burst 40 (token bucket at web)	2000	—	List deals
GET	/v1/deals/{idOrCode}	staff	all	safe		20/s per user, burst 40 (token bucket at web)	2000	—	Get a deal
PATCH	/v1/deals/{idOrCode}	staff	Admin, Manager, Demand agent	If-Match		20/s per user, burst 40 (token bucket at web)	2000	deal.updated.v1, deal.closed.v1, offer.commercial_status_changed.v1, demand.status_changed.v1, audit.recorded.v1	Advance stage, update terms / next action / follow-up; stage=Closed closes the deal (saga start)
POST	/v1/deals/{idOrCode}/cancel	staff	Admin, Manager, Demand agent	Idempotency-Key		20/s per user, burst 40 (token bucket at web)	2000	deal.cancelled.v1, deal.updated.v1, offer.commercial_status_changed.v1, demand.status_changed.v1, lifecycle.stage_changed.v1, audit.recorded.v1	Cancel a deal (compensation): offer → Available, demand → Active; deal kept
POST	/v1/deals/{idOrCode}/follow-ups	staff	Admin, Manager, Demand agent, Supply agent	Idempotency-Key		20/s per user, burst 40 (token bucket at web)	2000	deal.updated.v1, demand.confirmed.v1, offer.confirmed.v1, lifecycle.stage_changed.v1	Log a follow-up on an open deal and set the next one (resets both life curves)
GET	/v1/lease-renewals	staff	all	safe	yes	20/s per user, burst 40 (token bucket at web)	2000	—	Scheduled lease renewals (month 10 of 11-month leases, US-16)
GET	/v1/notifications	staff	all	safe	yes	20/s per user, burst 40 (token bucket at web)	2000	—	My in-app notifications (FR-NTF-1)
GET	/v1/notifications/unread-count	staff	all	safe		20/s per user, burst 40 (token bucket at web)	2000	—	Unread notification count (sidebar badge)
POST	/v1/notifications/mark-read	staff	all	Idempotency-Key		20/s per user, burst 40 (token bucket at web)	2000	—	Mark notifications read (by ids or up to a time)
GET	/v1/watchlist-tasks	staff	all	safe	yes	20/s per user, burst 40 (token bucket at web)	2000	—	Supply follow-up tasks for Market Signals (D-14, US-36)
PATCH	/v1/watchlist-tasks/{taskId}	staff	Admin, Manager	If-Match		20/s per user, burst 40 (token bucket at web)	2000	—	Assign, set due date or cancel a watchlist task
POST	/v1/watchlist-tasks/{taskId}/complete	staff	Admin, Manager, Supply agent	Idempotency-Key		20/s per user, burst 40 (token bucket at web)	2000	watchlist_task.completed.v1	Close a watchlist follow-up task

Method	Path	Auth	Roles	Idempotency	Paged	Rate limit	Timeout ms	Emits	Summary
GET	/v1/settings/life-curve-thresholds	staff	all	safe		20/s per user, burst 40 (token bucket at web)	2000	—	Life-curve thresholds per category
PUT	/v1/settings/life-curve-thresholds	staff	Admin	by design		20/s per user, burst 40 (token bucket at web)	2000	audit.recorded.v1	Replace life-curve thresholds (Admin); recomputes nextChangeOn
GET	/v1/settings/queue-weights	staff	all	safe		20/s per user, burst 40 (token bucket at web)	2000	—	Should call ranking weights
PUT	/v1/settings/queue-weights	staff	Admin	by design		20/s per user, burst 40 (token bucket at web)	2000	audit.recorded.v1	Replace Should call ranking weights (Admin)
GET	/internal/v1/subject-states	svc	—	safe	yes	10/s per calling service	2000	—	Life stage and Commercial axis per subject, for consumers rebuilding a projection
POST	/internal/v1/relay	cron	—	by design		1 call/min per job (pg_cron); internal only	10000	—	Relay unpublished outbox rows into consumer pgmq queues
POST	/internal/v1/drain/{queue}	cron	—	by design		1 call/min per job (pg_cron); internal only	55000	offer.commercial_status_changed.v1, demand.status_changed.v1, lifecycle.stage_changed.v1, sourcing_request.updated.v1	Drain a pgmq queue owned by this service
POST	/internal/v1/jobs/{name}	cron	—	by design		1 call/min per job (pg_cron); internal only	55000	queue.counts_changed.v1, deal.updated.v1, lifecycle.stage_changed.v1, demand.exited.v1, offer.commercial_status_changed.v1, lease_renewal.due.v1	Run a scheduled job (one bounded batch; continues on the next call)
GET	/health/live	none	—	safe		unlimited (platform probe)	500	—	Liveness: process is up
GET	/health/ready	none	—	safe		unlimited (platform probe)	1000	—	Readiness: DB reachable and migrations at the expected version

crm-engine (19)

Method	Path	Auth	Roles	Idempotency	Paged	Rate limit	Timeout ms	Emits	Summary
GET	/v1/demands/{idOrCode}/matches	staff	all	safe	yes	20/s per user, burst 40 (token bucket at web)	2000	—	Matches for a demand, best first (C-10, P-03)
GET	/v1/offers/{idOrCode}/matches	staff	all	safe	yes	20/s per user, burst 40 (token bucket at web)	2000	—	Matches that include an offer (single or bundle) (P-02)

Method	Path	Auth	Roles	Idempotency	Paged	Rate limit	Timeout ms	Emits	Summary
GET	/v1/matches/{idOrCode}	staff	all	safe		20/s per user, burst 40 (token bucket at web)	2000	—	Get a match
GET	/v1/matches/{idOrCode}/explanation	staff	all	safe		20/s per user, burst 40 (token bucket at web)	2000	—	Explain a match: hard filters, factor breakdown, flags
POST	/v1/matches/{idOrCode}/confirm	staff	Admin, Manager, Demand agent	Idempotency-Key		20/s per user, burst 40 (token bucket at web)	2000	match.confirmed.v1	Confirm a suggested match (US-21)
POST	/v1/matches/{idOrCode}/reject	staff	Admin, Manager, Demand agent	Idempotency-Key		20/s per user, burst 40 (token bucket at web)	2000	match.rejected.v1	Reject a match with a reason code (US-21)
POST	/v1/bundles	staff	Admin, Manager, Demand agent, Supply agent	Idempotency-Key		20/s per user, burst 40 (token bucket at web)	2000	match.suggested.v1, match.confirmed.v1	Build a bundle of 2–3 offers for a demand (A-38)
GET	/v1/bundles/{idOrCode}	staff	all	safe		20/s per user, burst 40 (token bucket at web)	2000	—	Get a bundle
GET	/v1/demands/{idOrCode}/exclusions	staff	all	safe	yes	20/s per user, burst 40 (token bucket at web)	2000	—	Offers excluded for a demand, with reasons (date-aware, e.g. 'Available too late')
POST	/v1/demands/{idOrCode}/matching-runs	staff	Admin, Manager, Demand agent	Idempotency-Key		20/s per user, burst 40 (token bucket at web)	2000	demand.matching_completed.v1, match.suggested.v1, match.closed.v1, match.flagged.v1	Re-run matching for one demand (async)
GET	/v1/matching-runs/{runId}	staff	all	safe		20/s per user, burst 40 (token bucket at web)	2000	—	Status of a matching run
GET	/v1/weights	staff	all	safe		20/s per user, burst 40 (token bucket at web)	2000	—	Current match weights and tunables
PUT	/v1/weights	staff	Admin	by design		20/s per user, burst 40 (token bucket at web)	2000	audit.recorded.v1, match.suggested.v1, match.closed.v1	Replace match weights (Admin); queues a full re-score
GET	/internal/v1/matches	svc	—	safe	yes	10/s per calling service	2000	—	All matches (keyset), for consumers rebuilding a projection
POST	/internal/v1/relay	cron	—	by design		1 call/min per job (pg_cron); internal only	10000	—	Relay unpublished outbox rows into consumer pgmq queues

Method	Path	Auth	Roles	Idempotency	Paged	Rate limit	Timeout ms	Emits	Summary
POST	/internal/v1/drain/{queue}	cron	—	by design		1 call/min per job (pg_cron); internal only	55000	match.suggested.v1, match.closed.v1, match.flagged.v1, match.reopened.v1, demand.matching_completed.v1	Drain a pgmq queue owned by this service
POST	/internal/v1/jobs/{name}	cron	—	by design		1 call/min per job (pg_cron); internal only	55000	match.suggested.v1, match.closed.v1, match.flagged.v1, match.reopened.v1, demand.matching_completed.v1	Run a scheduled job (one bounded batch; continues on the next call)
GET	/health/live	none	—	safe		unlimited (platform probe)	500	—	Liveness: process is up
GET	/health/ready	none	—	safe		unlimited (platform probe)	1000	—	Readiness: DB reachable and migrations at the expected version

listings (25)

Method	Path	Auth	Roles	Idempotency	Paged	Rate limit	Timeout ms	Emits	Summary
GET	/v1/offers/{idOrCode}/publication	staff	all	safe		per user at web (20 req/s, burst 40)	2000	—	Publication state of an offer with the computed ceiling and the reasons (C-12, P-02 Overview)
PUT	/v1/offers/{idOrCode}/publication	staff	Admin, Manager, Supply agent	by design		per user at web (20 req/s, burst 40)	2000	publication.changed.v1, audit.recorded.v1	Set the publication level (at or below the ceiling), optionally with an edited public description
POST	/v1/offers/{idOrCode}/privacy-scan	staff	Admin, Manager, Supply agent	Idempotency-Key		per user at web (20 req/s, burst 40)	2000	—	Dry-run privacy scan of a proposed public description (and the offer's photos) without changing the level
GET	/v1/projects/{idOrCode}/publication	staff	all	safe		per user at web (20 req/s, burst 40)	2000	—	Publication state of a project (P-05) with ceiling and reasons
PUT	/v1/projects/{idOrCode}/publication	staff	Admin, Manager, Supply agent	by design		per user at web (20 req/s, burst 40)	2000	publication.changed.v1, audit.recorded.v1	Set a project's level (Private or Public; projects have no Anonymous level)
GET	/v1/demands/{idOrCode}/demand-post	staff	all	safe		per user at web (20 req/s, burst 40)	2000	—	Anonymous demand post state for a demand (P-03 Sourcing tab)
PUT	/v1/demands/{idOrCode}/demand-post	staff	Admin, Manager, Demand agent	by design		per user at web (20 req/s, burst 40)	2000	publication.changed.v1, audit.recorded.v1	Post (Anonymous) or take down (Private) the anonymous demand post; allowed only while the demand is in Sourcing and Fresh/Ageing
GET	/v1/publications	staff	all	safe	yes	per user at web (20 req/s, burst 40)	2000	—	Staff list of publication states (e.g. "Stale Public offers", everything Public), stored-field filters

Method	Path	Auth	Roles	Idempotency	Paged	Rate limit	Timeout ms	Emits	Summary
GET	/v1/publication-settings	svc, staff	all	safe		per user at web (20 req/s, burst 40)	2000	—	RERA agent number and listing notes shown on every item
PUT	/v1/publication-settings	staff	Admin	by design		per user at web (20 req/s, burst 40)	2000	audit.recorded.v1	Set the 11 Estates MahaRERA agent number and the "subject to confirmation" note (Admin)
GET	/v1/api-keys	staff	Admin	safe	yes	per user at web (20 req/s, burst 40)	2000	—	List website API keys (metadata only; secrets are never returned after creation)
POST	/v1/api-keys	staff	Admin	Idempotency-Key		per user at web (20 req/s, burst 40)	2000	audit.recorded.v1	Create an API key for a website or microsite; the secret is returned once
POST	/v1/api-keys/{keyId}/rotate	staff	Admin	Idempotency-Key		per user at web (20 req/s, burst 40)	2000	audit.recorded.v1	Issue a replacement key; the old key stays valid for a grace period (default 168 h, max 720 h)
POST	/v1/api-keys/{keyId}/revoke	staff	Admin	Idempotency-Key		per user at web (20 req/s, burst 40)	2000	audit.recorded.v1	Revoke a key immediately (the edge cache entries for it expire within 30 s)
GET	/v1/listings	key	Website client	safe	yes	50 req/s, burst 100 per API key (token bucket enforced by listings itself, conventions R-1; cache hits are not counted)	2000	—	Published offers (Anonymous or Public shape per item), stored-field filters, cursor pagination (max 50)
GET	/v1/listings/{publicId}	key	Website client	safe		50 req/s, burst 100 per API key	2000	—	One published offer. Withdrawn or unknown IDs → 404 (the change feed says "withdrawn").
GET	/v1/projects	key	Website client	safe	yes	50 req/s, burst 100 per API key	2000	—	Published projects (Sale, Primary)
GET	/v1/projects/{publicId}	key	Website client	safe		50 req/s, burst 100 per API key	2000	—	One published project with configurations
GET	/v1/demand-posts	key	Website client	safe	yes	50 req/s, burst 100 per API key	2000	—	Anonymous demand posts ("Wants" label) while the demand is in Sourcing
GET	/v1/changes	key	Website client	safe		50 req/s, burst 100 per API key	2000	—	Change feed (published / updated / upgraded / downgraded / withdrawn) since a cursor or a time
POST	/internal/v1/relay	cron	scheduler	by design		none (scheduler only; single-flight via advisory lock)	55000	publication.changed.v1, audit.recorded.v1	Publish unpublished outbox rows to subscribed consumer queues (pg_cron every minute + poke after commit)
POST	/internal/v1/drain/{queue}	cron	scheduler	by design		none (scheduler only; every 20 s + poke from the producer relay)	55000	publication.changed.v1, audit.recorded.v1	Consume a batch from listings' queue (dedupe on eventId, per-aggregate version check, DLQ after 5 attempts)
POST	/internal/v1/jobs/{name}	cron	scheduler	by design		none (scheduler only; single-flight per job name)	55000	publication.changed.v1	Scheduled jobs
GET	/health/live	none	anonymous	safe		none	500	—	
GET	/health/ready	none	anonymous	safe		none	1000	—	

insight (20)

Method	Path	Auth	Roles	Idempotency	Paged	Rate limit	Timeout ms	Emits	Summary
GET	/v1/chat/conversations	staff	all	safe	yes	per user at web (20 req/s, burst 40)	2000	—	The caller's own conversations (sidebar "Recent chats"), newest first
POST	/v1/chat/conversations	staff	all	Idempotency-Key		per user at web (20 req/s, burst 40)	2000	—	Start a conversation (" + New chat")
GET	/v1/chat/conversations/{conversationId}	staff	all	safe		per user at web (20 req/s, burst 40)	2000	—	One of the caller's conversations (metadata)
DELETE	/v1/chat/conversations/{conversationId}	staff	all	by design		per user at web (20 req/s, burst 40)	2000	—	Delete one of the caller's conversations (soft delete now, purged by the nightly job)
GET	/v1/chat/conversations/{conversationId}/messages	staff	all	safe	yes	per user at web (20 req/s, burst 40)	2000	—	Messages of a conversation, oldest first (stored text is redacted; cards are re-rendered with live state by the UI, R-CHAT-3)
POST	/v1/chat/conversations/{conversationId}/messages	staff	all	Idempotency-Key		at web: 1 concurrent stream and 30 messages/min per user	15000	—	Ask a question; the answer streams back as Server-Sent Events
GET	/v1/chat/plan-catalogue	staff	all	safe		per user at web (20 req/s, burst 40)	2000	—	The allowed query-plan templates (what the assistant can answer), for transparency and the UI's "How I got this"
POST	/v1/queries	staff	all	by design		per user at web (20 req/s, burst 40)	2000	—	Run a validated plan (no model involved) and return one page of rows — used by "Open in panel" (P-07) and dashboard drill-downs
GET	/v1/dashboards/demand	staff	Admin, Manager, Demand agent, Supply agent	safe		per user at web (20 req/s, burst 40)	2000	—	Demand dashboard (BRD §8): by source, life curve, team queues, exits, classification grid with Wants labels
GET	/v1/dashboards/supply	staff	Admin, Manager, Demand agent, Supply agent	safe		per user at web (20 req/s, burst 40)	2000	—	Supply dashboard (BRD §8): stock, life curve, queues and listings, classification grid with For labels, deal-tag filters
GET	/v1/dashboards/scopes	staff	Admin, Manager, Demand agent, Supply agent	safe		per user at web (20 req/s, burst 40)	2000	—	Other scopes (BRD §8): Business and Capital desks, Archive, Network, Watchlist
GET	/v1/dashboards/quality	staff	all	safe		per user at web (20 req/s, burst 40)	2000	—	Data quality (BRD §8): uploads, review queue by review_reason, side checks, source quality
GET	/v1/exports	staff	all	safe	yes	per user at web (20 req/s, burst 40)	2000	—	The caller's exports (Admin sees all), newest first
POST	/v1/exports	staff	all	Idempotency-Key		per user at web (20 req/s, burst 40); plus 10 exports per hour per user (insight)	2000	export.completed.v1, export.failed.v1, audit.recorded.v1	Start a background Excel export of a list or chat answer (≤ 100,000 rows; pilot ≤ 20,000, R-16)

Method	Path	Auth	Roles	Idempotency	Paged	Rate limit	Timeout ms	Emits	Summary
GET	/v1/exports/{idOrCode}	staff	all	safe		per user at web (20 req/s, burst 40)	2000	—	Status of an export; when completed and not expired, a short-lived signed download URL (file kept 24 h)
POST	/internal/v1/relay	cron	scheduler	by design		none (scheduler only; single-flight via advisory lock)	55000	export.completed.v1, export.failed.v1, audit.recorded.v1	Publish outbox rows to consumer queues (pg_cron every minute + poke after commit)
POST	/internal/v1/drain/{queue}	cron	scheduler	by design		none (scheduler only; every minute + poke)	55000	export.completed.v1, export.failed.v1, audit.recorded.v1	Consume a batch: q_insight (domain events → read model), q_insight_exports (export jobs)
POST	/internal/v1/jobs/{name}	cron	scheduler	by design		none (scheduler only; single-flight per job)	55000	—	Scheduled jobs
GET	/health/live	none	anonymous	safe		none	500	—	
GET	/health/ready	none	anonymous	safe		none	1000	—	